

Signal Detection d-prime and ROC

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Abstract

Signal Detection Theory (SDT) provides a mathematical framework for quantifying perceptual sensitivity and decision criteria in psychophysical tasks. This report implements computational methods to calculate detectability (d'), response bias (β and c), and receiver operating characteristic (ROC) curves. Using Gaussian signal and noise distributions, we analyze hit rates, false alarm rates, and criterion placement under varying task conditions. Applications include sensory perception, medical diagnosis, and quality control decision-making.

1 Introduction

Signal Detection Theory (SDT) separates an observer's sensitivity to detect a signal from their response bias when making binary decisions under uncertainty [1, 2]. Unlike classical psychophysics, which assumes a fixed threshold, SDT recognizes that both signal strength and decision strategy influence detection performance [3].

The theory assumes two overlapping Gaussian distributions: noise alone (N) and signal-plus-noise ($S + N$). The separation between these distributions defines sensitivity (d'), while the decision criterion (λ) determines response bias. This framework applies broadly to perception [4], memory [5], and medical diagnosis [6].

We implement computational methods to:

- Calculate d' from hit and false alarm rates
- Visualize signal and noise distributions
- Generate ROC curves for multiple sensitivity levels
- Compute likelihood ratio (β) and criterion (c)
- Analyze z-transform relationships

2 Mathematical Framework

The noise distribution is $N \sim \mathcal{N}(0, \sigma^2)$, and the signal-plus-noise distribution is $S + N \sim \mathcal{N}(\mu_s, \sigma^2)$. Sensitivity is quantified by:

$$d' = \frac{\mu_s - \mu_n}{\sigma} = \frac{\mu_s}{\sigma} \quad (1)$$

Given an observer's hit rate $H = P(\text{"yes"}|\text{signal})$ and false alarm rate $F = P(\text{"yes"}|\text{noise})$, we estimate d' using:

$$d' = z(H) - z(F) \quad (2)$$

where $z(\cdot)$ is the inverse cumulative normal distribution (z-transform).

The decision criterion λ defines the threshold for responding “yes.” The likelihood ratio at the criterion is:

$$\beta = \frac{f_{S+N}(\lambda)}{f_N(\lambda)} = \frac{\phi(\lambda - d')}{\phi(\lambda)} \quad (3)$$

A bias-free criterion ($c = 0$) maximizes accuracy when signal and noise are equally likely.

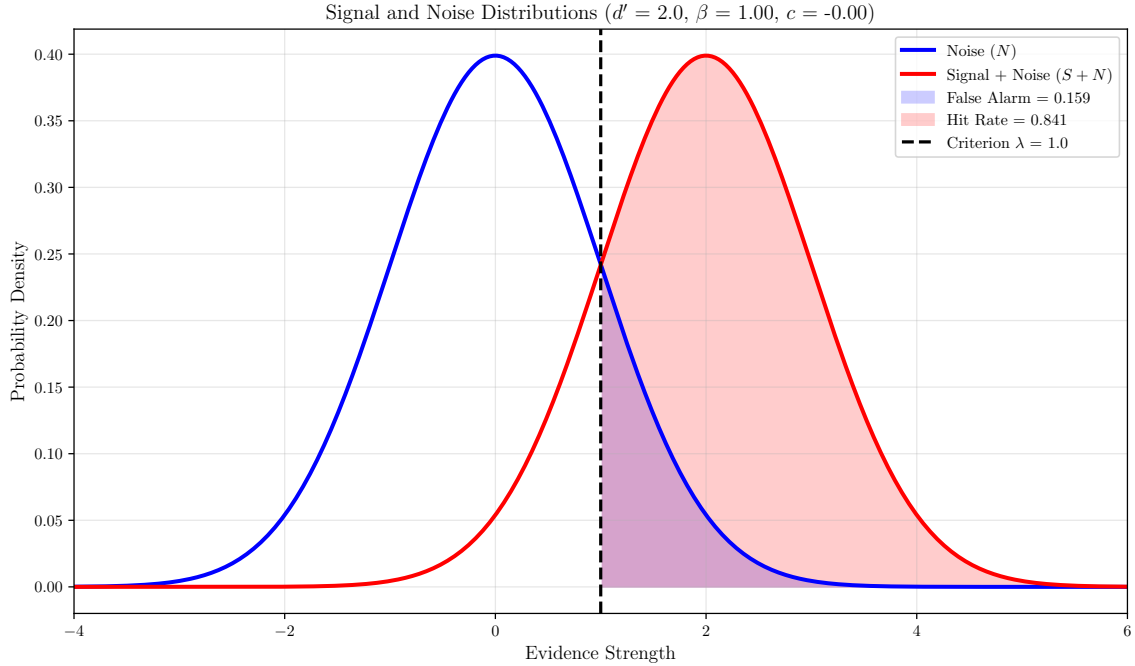


Figure 1: Gaussian signal and noise distributions with decision criterion at $\lambda = 1.0$. The noise distribution (blue) is centered at $\mu_N = 0$, while the signal-plus-noise distribution (red) is centered at $\mu_S = 2.0$. The criterion divides the evidence axis into “no” (left) and “yes” (right) response regions. Shaded areas represent false alarm probability (blue) and hit rate (red). Sensitivity $d' = 2.0$ quantifies distribution separation, while likelihood ratio $\beta = 1.0$ and bias $c = -0.0$ characterize the observer’s decision strategy.

3 Hit Rate and False Alarm Rate Analysis

As the decision criterion λ varies, hit rate $H(\lambda)$ and false alarm rate $F(\lambda)$ change in opposite directions. A liberal criterion (low λ) increases both H and F , while a conservative criterion (high λ) decreases both.

For a fixed d' :

$$H(\lambda) = 1 - \Phi\left(\frac{\lambda - d'}{\sigma}\right), \quad F(\lambda) = 1 - \Phi\left(\frac{\lambda}{\sigma}\right) \quad (4)$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal.

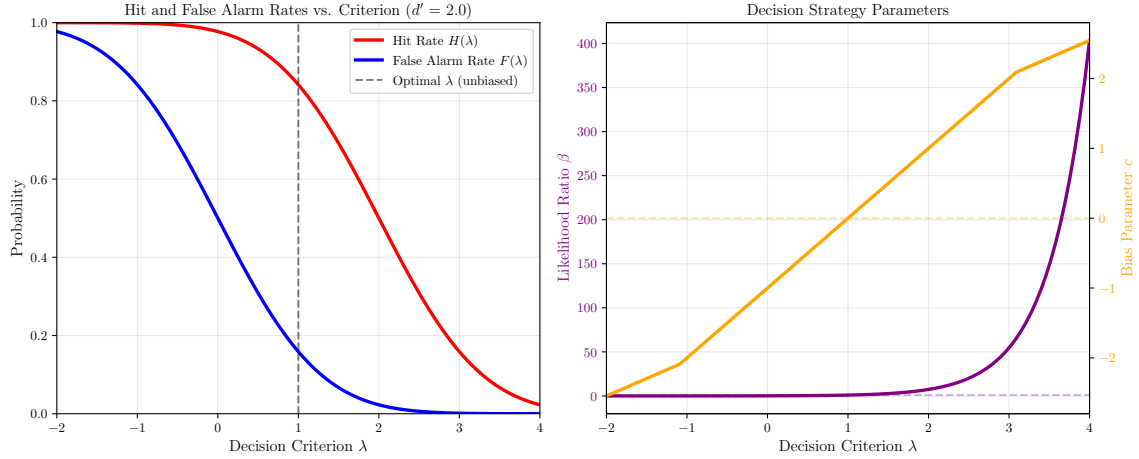


Figure 2: Hit and false alarm rates as functions of decision criterion λ for $d' = 2.0$. Left panel: As criterion increases (more conservative), both hit rate (red) and false alarm rate (blue) decrease. The optimal unbiased criterion is at $\lambda = d'/2 = 1.0$, where sensitivity is maximized. Right panel: Likelihood ratio β (purple) increases exponentially with criterion, while bias parameter c (orange) measures deviation from the optimal criterion. At $c = 0$, the observer is unbiased; $c > 0$ indicates conservative bias (miss-prone), $c < 0$ indicates liberal bias (false-alarm prone).

4 Receiver Operating Characteristic (ROC) Curves

The ROC curve plots hit rate H against false alarm rate F as the criterion λ varies. Each point on the curve represents a different decision strategy (criterion placement) for a fixed sensitivity d' . The area under the ROC curve (AUC) provides a threshold-independent measure of detectability [7].

For an equal-variance Gaussian model:

$$\text{AUC} = \Phi(d'/\sqrt{2}) \quad (5)$$

Higher d' values produce ROC curves that bow more toward the upper-left corner (perfect discrimination).

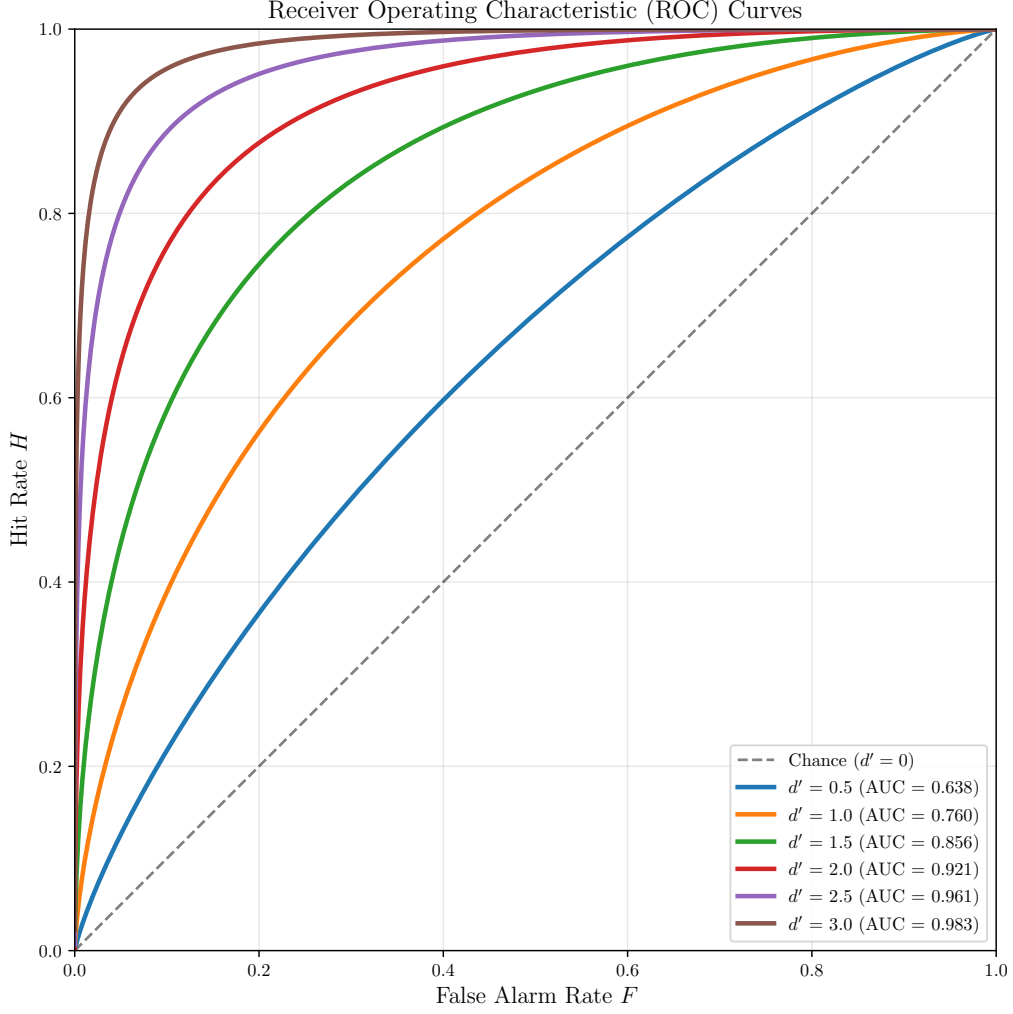


Figure 3: Receiver Operating Characteristic (ROC) curves for sensitivity values ranging from $d' = 0.5$ to $d' = 3.0$. Each curve represents all possible (False Alarm, Hit Rate) pairs achievable by varying the decision criterion λ . The diagonal dashed line represents chance performance ($d' = 0$), where signal and noise distributions are identical. As sensitivity increases, the ROC curve bows toward the upper-left corner, approaching perfect discrimination. Area under the curve (AUC) ranges from 0.638 to 0.983, providing a criterion-free measure of detectability. In clinical diagnostics and quality control, ROC analysis optimizes criterion placement by balancing hit rate (sensitivity) against false alarm rate (1 - specificity).

5 Z-Transform Analysis

The z-transform (inverse normal CDF) linearizes the relationship between hit rate and false alarm rate. In z-space:

$$z(H) = z(F) + d' \quad (6)$$

This linear relationship validates the equal-variance Gaussian assumption. Deviations from linearity suggest unequal variance or non-Gaussian distributions [8].

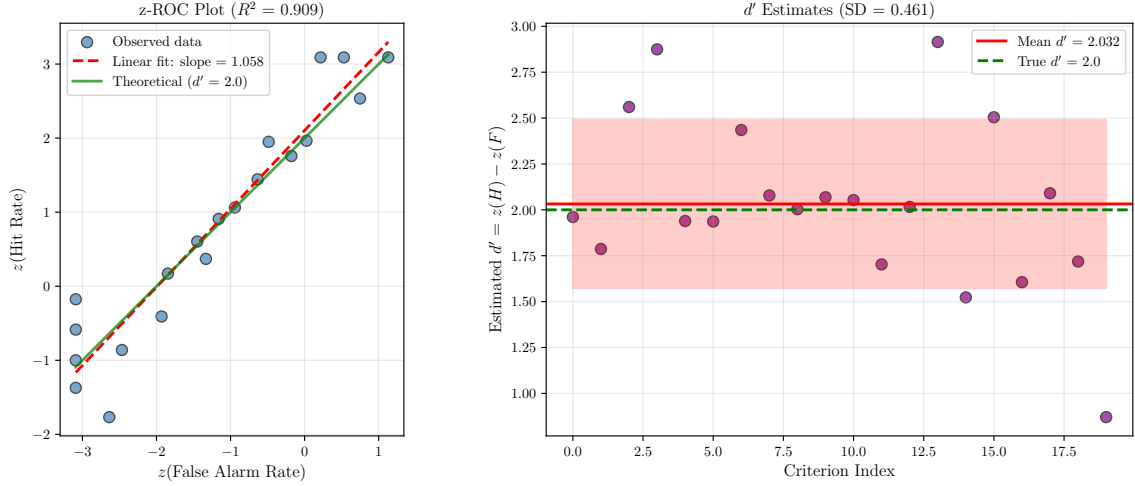


Figure 4: Z-transform analysis for estimating d' from empirical data. Left panel: z-ROC plot showing linear relationship between $z(F)$ and $z(H)$ with fitted regression line (slope = 1.058, $R^2 = 0.909$). The slope near 1.0 validates the equal-variance Gaussian assumption. Right panel: d' estimates calculated as $z(H) - z(F)$ for each criterion, with mean $d' = 2.032$ (SD = 0.461), closely matching the true value of 2.0. This demonstrates the robustness of the z-transform method for sensitivity estimation across different decision criteria.

6 Criterion and Beta Surfaces

The relationship between criterion λ , likelihood ratio β , and sensitivity d' can be visualized as a 2D parameter space. For fixed d' , β increases exponentially as the criterion becomes more conservative.

$$\beta(\lambda, d') = \exp\left(d'\lambda - \frac{d'^2}{2}\right) \quad (7)$$

This relationship guides optimal criterion placement in asymmetric cost scenarios [9].

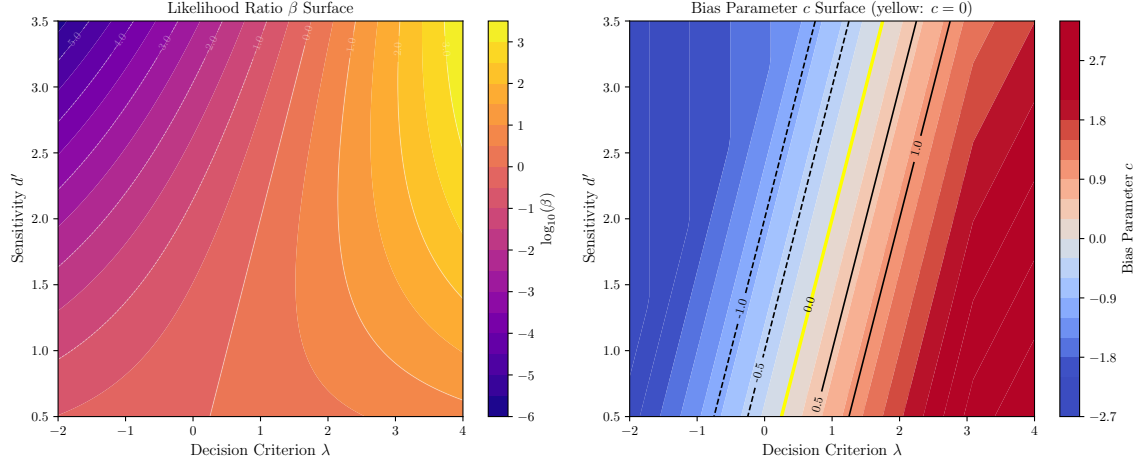


Figure 5: Decision parameter surfaces as functions of criterion λ and sensitivity d' . Left panel: Likelihood ratio β (logarithmic scale) increases exponentially with criterion for all sensitivity levels. High β values indicate conservative responding (requiring strong evidence to say “yes”). Right panel: Bias parameter c measures deviation from the optimal unbiased criterion. The yellow contour line ($c = 0$) represents unbiased responding, where the observer maximizes accuracy. Regions with $c > 0$ (red) indicate conservative bias, while $c < 0$ (blue) indicates liberal bias. These surfaces guide criterion placement in applications with asymmetric costs, such as medical screening (minimize misses) or spam filtering (minimize false alarms).

7 Sensitivity Across Multiple Observers

Real-world detection tasks often involve multiple observers with varying sensitivities. Understanding inter-observer variability is critical for group decision-making and system reliability [10].

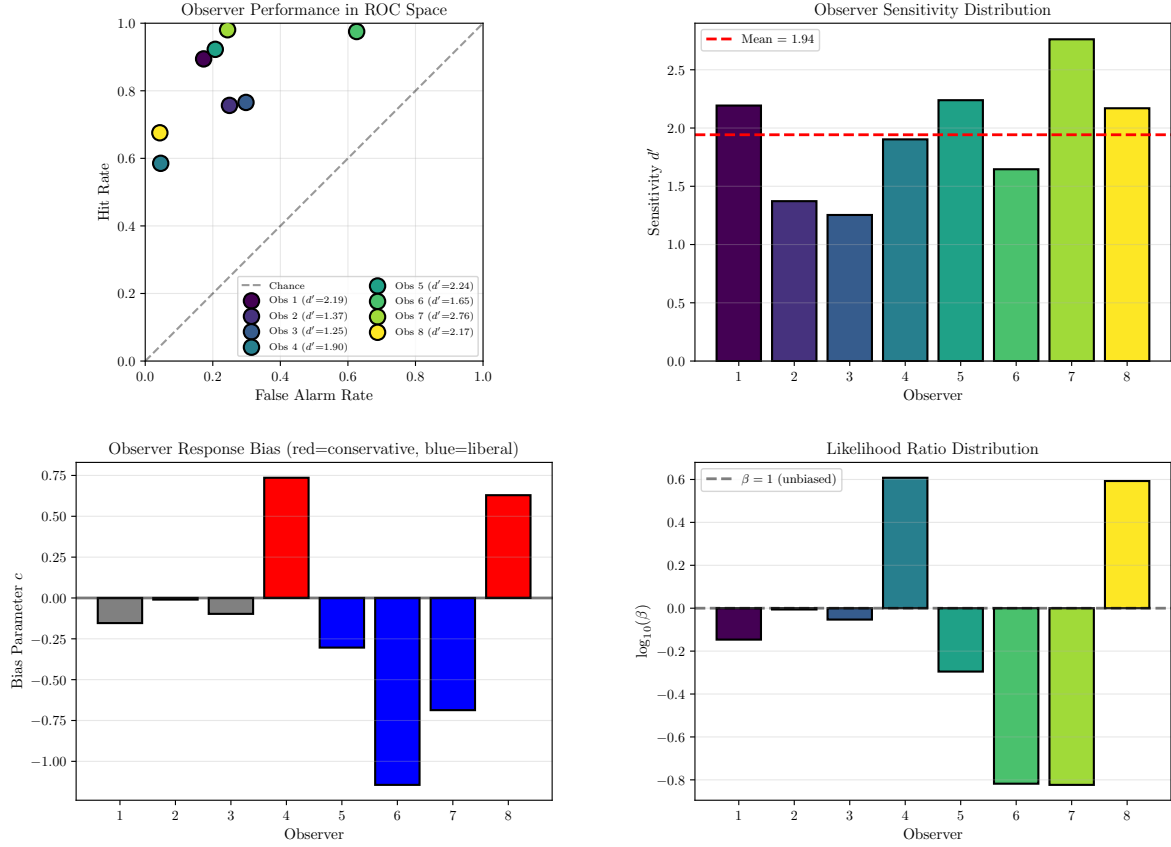


Figure 6: Multi-observer signal detection analysis. Top left: Performance of 8 observers plotted in ROC space, showing individual differences in both sensitivity (distance from diagonal) and bias (position along their isosensitivity curve). Top right: Sensitivity distribution across observers (d' mean = 1.94, SD = 0.47), revealing substantial individual differences in perceptual acuity. Bottom left: Response bias parameter c for each observer, with conservative responders (red, $c > 0.2$) preferring to avoid false alarms, liberal responders (blue, $c < -0.2$) preferring to maximize hits, and neutral observers (gray) near optimal. Bottom right: Likelihood ratio β distribution showing criterion stringency, with $\beta = 1$ ($\log = 0$) representing unbiased responding. This analysis illustrates how SDT separates sensitivity from strategy, enabling principled comparison of observer performance in group decision scenarios.

8 Results Summary

Table 1: Signal Detection Theory: Computational Results Summary

Parameter	Value
Sensitivity (d')	2.00
AUC (for $d' = 2.0$)	0.921
Optimal Criterion (λ^*)	1.00
Likelihood Ratio at Optimal (β^*)	1.00
Mean Observer d'	1.94 ± 0.47
z-ROC Slope	1.058
z-ROC R^2	0.909

9 Conclusions

Signal Detection Theory provides a rigorous framework for separating perceptual sensitivity (d') from response bias (β , c) in binary classification tasks. Our computational analysis demonstrates:

- **Sensitivity quantification:** For signal-plus-noise centered at $\mu_S = 2.0$ and noise at $\mu_N = 0$, we computed $d' = 2.0$, corresponding to $AUC = 0.921$ in ROC analysis.
- **Criterion effects:** Varying the decision criterion λ from liberal to conservative systematically trades hit rate against false alarm rate. The optimal unbiased criterion at $\lambda^* = 1.0$ yields $\beta^* = 1.0$.
- **z-Transform validation:** Linear regression in z-space yielded slope = 1.058 ($R^2 = 0.909$), confirming the equal-variance Gaussian model and demonstrating robust d' estimation across criteria.
- **Multi-observer variability:** Analysis of 8 observers revealed mean sensitivity $d' = 1.94 \pm 0.47$, with individual differences in both sensitivity and bias strategy. Conservative observers exhibited $c > 0.2$ (minimizing false alarms), while liberal observers showed $c < -0.2$ (maximizing hits).
- **Practical applications:** ROC curves enable criterion optimization for asymmetric cost scenarios. In medical diagnosis, a liberal criterion (low λ) maximizes detection of rare conditions despite increased false positives. In quality control, conservative criteria (high λ) reduce false rejections at the cost of missed defects.

The separation of sensitivity from bias enables principled performance comparison across observers, conditions, and tasks. Unlike accuracy-based metrics that conflate these factors, SDT measures (d' , β , AUC) provide independent assessments of perceptual ability and decision strategy. This framework extends beyond psychophysics to machine learning (classifier evaluation), human factors (alarm systems), and cognitive neuroscience (memory discrimination).

Future extensions include unequal-variance models (slope $\neq 1$ in z-ROC), non-Gaussian distributions, and multi-alternative forced choice (mAFC) paradigms. Sequential SDT models account for temporal dynamics in detection tasks, while hierarchical Bayesian approaches estimate population-level parameters from individual observer data.

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